Shreya Abraham Varghese

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Summary

A Full-Stack AI Engineer with experience building end-to-end applications using Python, JavaScript, APIs, databases, and LLM-based workflows. Developed production-oriented solutions including document Q&A systems, website lead-capture pipelines, workflow automation tools, and machine-learning applications. Comfortable translating operational requirements into practical technical solutions, working across frontend, backend, data pipelines, and deployment workflows.

Profiles

Github
sv8911

LinkedIn
Shreya Abraham Varghese

Experience

Krishna Kartik Homes — September 2024 - Present
Junior Python Developer

- Developing and maintaining a responsive real-estate website using HTML, CSS, and JavaScript, with reusable property pages, enquiry forms, popup flows, and interactive tools.
- Building a backend lead-capture workflow using API endpoints and database storage to validate, store, and manage enquiries submitted through multiple website forms.
- Implemented a RAG-based document Q&A system using Python, embeddings, and semantic search, enabling retrieval across 1,000+ document chunks.
- Developed a lead-classification model to categorise enquiries as Hot, Warm, or Cold, supporting prioritised follow-ups.
- Built internal tools including an EMI calculator and property-tax estimator to support sales conversations.

ServiceNow — Jan 2026 – April 2026
Trainee – Rise Up with ServiceNow (AuraSync Project)

- Built an employee lifecycle management application automating leave delegation, internal transfers, and offboarding workflows.
- Developed custom tables, UI policies, user roles, and access-control rules within a scoped ServiceNow application.
- Implemented multi-step Flow Designer workflows for approvals, task reassignment, and lifecycle tracking.
- Automated access provisioning and revocation processes to reduce manual effort and improve operational consistency.

CodeClause — August 2023 - September 2023
Machine Learning Intern

Built a real-time face detection pipeline using Python, OpenCV, and Flask, achieving 25–30 FPS and improving accuracy by 20% through preprocessing and multi-scale detection. Deployed via REST API for low-latency inference.

Languages

English	Hindi	Malayalam
Highly Proficient	Native	Proficient

Projects

Semi-Supervised Brain-Tumor Segmentation Using a Box-Constrained Loss in a 2D U-Net
Developed a lightweight MobileNetV2–U-Net pipeline for whole-tumor segmentation on MRI scans using a semi-supervised, mixed-annotation strategy. Designed a two-stage training process combining mask-supervised pre-training with bounding-box–guided fine-tuning. Introduced distance-map priors and outside-box penalties to learn from coarse labels, reducing reliance on costly voxel-level annotations. Delivered a reproducible pipeline with documented error analysis and scalability considerations.

A Multi-Model Approach to Relation Extraction: BERT Embeddings with XGBoost and Graph Neural Networks
Designed a hybrid system for document-level relation extraction using BERT embeddings. Compared performance of XGBoost and Graph Convolutional Networks (GCNs) on the Re-DocRED dataset. XGBoost achieved superior generalization (Micro F1: 66.05%) while GCN captured more true relations with higher recall. Focused on multi-label classification, precision-recall optimization, and long-range dependency modeling

Advanced Neural Network Architecture for Brain Lesion Identification
Developed a deep learning system to classify brain tumors from MRI scans using VGG16, InceptionV3, and Xception models. Achieved up to 98% accuracy, with VGG16 performing best. Processed a dataset of 3,000 images and conducted comparative analysis to optimize model performance. Improved diagnostic speed and accuracy, demonstrating potential for clinical application.

Skills

Languages: Python, SQL

AI/ML: Machine Learning, Deep Learning, Neural Networks, Computer Vision, NLP

LLM & Data Systems: Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search

Backend & APIs: Flask, REST APIs

Tools & Platforms: ServiceNow, Git, Google Sheets API

Education

The University of Manchester — Masters of Science
Advanced computer science, Artificial Intelligence — September 2024 - December 2025

SRM Institute of Science and Technology — Bachelors in Technology
Computer Science — September 2020 - May 2024

Certification

Fundamentals of Deep Learning
NVIDIA

Machine Learning with Python
IBM

Data Analysis with Python
IBM

Python Project for Data science
IBM

Power BI Virtual Case Experience
PWC

AWS Academy Cloud Operations
AWS

Database Foundations
Oracle